



Time Series Anomaly Detection with Variational Autoencoder Using Mahalanobis Distance

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Abstract. Two themes have dominated the research on anomaly detection in time series data, one related to explorations of deep architectures for the task, and the other, equally important, the creation of large benchmark datasets. In line with the current trends, we have proposed several deep learning architectures based on Variational Autoencoders that have been evaluated for detecting cyber-attacks on water distribution system on the BATADAL challenge task and dataset. The second research aim of this study was to examine the impact of using Mahalanobis distance as a reconstruction error on the performance of the proposed models.

Keywords: Time series analysis · Anomaly detection · Variational autoencoder · Mahalanobis distance

1 Introduction

The ability to draw valid inferences from data has critical importance for a number of applications and industrial sectors, from mobile networks to medical diagnosis to manufacturing control systems. Malicious attacks (e.g., intrusions, fraudulent behavior, cyber-attacks), abnormality (e.g., malignant cells in images, irregular system behavior) and faulty measurements (e.g., sensors, industrial equipment, medical devices) however, pose a serious threat to the validity of the assertion drawn from data. Detecting and labeling anomaly events that are sparse, context-specific, difficult to define or distinguished from what is considered normal data are far from straightforward. Addressing these longstanding concerns have led to further advances in anomaly detection that has been regarded as an active area of research that complements predictive and prescriptive analytics.

Research work on the topic expanded from traditional approaches relying on classification, distance-based and statistical modeling [1–11] to exploiting the advances of diverse deep learning architectures [12, 13]. New lines of research using deep generative neural networks [14, 15] are being hailed for their potential for unsupervised learning from multivariate data with spatial and temporal

dependencies. As with past imports in other areas, deep learning carries both great promise and substantial problems that are far from straightforward. From selecting the right deep architecture to learning its parameters, from scaling up the algorithms to understanding and interpreting model behaviors. These problems are faced by researchers and practitioners in the field that still awaits for robust and innovative guidelines and solutions.

Detecting cases of intrusion and anomaly in data observations entails 1) learning data representations of normal data i.e., mapping high-dimensional input data into a low-dimensional latent space, and 2) selecting the measure and threshold that will be used for distinguishing anomaly in the latent space. Handling high-dimensional data with spatial and temporal dependencies is pointed out as a major challenge for traditional approaches for anomaly detection, especially those inclined toward classification and clustering approaches. Principle Component Analysis (PCA) [11] is usually excluded, although the problem of hard-to-interpret anomaly data remains.

Extracting nonlinear and hierarchical discriminative features from complex high-dimensional data generated from IoT devices or real-time streaming data usually implies massive quantities of data that is sequential in nature and manifest characteristics such as trends and seasonality. Various deep learning architectures are suggested to handle the challenge. Convolutional Neural Networks (CNN) are considered better choice for capturing spatial and temporal dependencies when compared with the capabilities of Multilayer perceptron (MPA), while Recurrent neural networks (RNN) have been preferred for modeling sequence data. A special kind of RNNs, Long Short Term memory (LSTMs) are regarded particularly suited for capturing long-term dependencies in time-series data. Research on anomaly detection has established the central role deep learning plays in the success of a number of case studies [16–19]. Chalapathy and Chawla [13] present a comprehensive study of deep learning techniques for detecting anomalies and intrusions and highlights the performance advantages of deep learning models when faced with massive quantities of data.

Research interest in deep generative networks for anomaly detection, in particular Variational Autoencoders (VAE) [14], has been intense in recent years. Variational Autoencoders are particularly useful for generating encodings of input data and then use them to re-construct its input, which in the case of anomaly detection corresponds to denoting a data sample that deviates from what is considered normal as the one that cannot be reconstructed. In addition, the anomaly scores and decision thresholds are probabilistic with a statistical meaning.

Following these current trends in deep learning for anomaly detection [16–19], we have evaluated the performance of various deep learning models using Variational Autoencoders on the Battle of the Attack Detection Algorithms – BATADAL¹ challenge task, which was introduced and extensively described in [20]. The second objective of this paper was to investigate the potential

¹ <https://www.batadal.net/data.html>.

performance gains if Mahalanobis distance measure [21] was used instead of mean square error for detecting data that deviates from normal.

In the next section, we provide a brief historical overview of traditional methods comparing them to the current trends in anomaly detection. We then give a description of the BATADAL dataset used in our experiments. In the fourth section, the Variational Autoencoders are presented, which is a basic component in the proposed deep models, followed by a discussion of the ten hierarchical deep architectures that were evaluated on the task at hand. We assess the current and potential contributions of the proposed models in the fifth section. The final sections concludes the paper and proposes a research agenda for future explorations.

2 Related Work

Significant number of methods have been explored over the last decades indicating the longstanding interest in the field of anomaly detection. In this short overview, we place emphasis on recent studies using deep learning for anomaly detection, but we also want to give reference to traditional approaches that still have a place in today’s research.

Early work on anomaly detection tend to originate from machine learning standpoint toward detecting anomalies. Supervised or semi-supervised classification methods such as SVM [1], S3VM [2] or decision tree based models [3–5] such as Random Forrest [4, 5] were used to distinguish between normal and outlier class. Furthermore, unsupervised methods such as proximity-based methods [6–8] (e.g. Nearest Neighbor [6], K-means [6], Lof [8]), angle-based methods [9], Isolation Forrest [10] or PCA [11] were used to separate normal from abnormal data. The premise in statistical modeling is that normal data occur in high-probability regions of the selected distribution (e.g., Gaussian Mixture Model) that closely resembles the realistic data distribution. The major limitations of the traditional methods is their inability to handle massive quantities of complex high-dimensional data including those with dependencies.

Consistent with the accomplishments of deep learning for predictive analysis in a number of areas, from image processing to language understanding, various hierarchical deep neural networks have been used to learn nonlinear and discriminative indicators in multi-dimensional data that facilitates the task of anomaly detection. Two types of deep learning for anomaly detection are distinguished, prediction and reconstruction based. Prediction inclined ones are used to predict a point in future and detect anomaly based on the difference between the actual and predicted value i.e., prediction error. Different contexts and applications are likely to afford different neural architectures. DeepNap, Kim et al. (2018) [16] uses an AE with LSTM for prediction, and fully-connected MLP for detecting an anomaly in time-series data, using partial reconstruction error to train both modules jointly, while LSTM neural networks and multivariate Gaussian distribution as a prediction error have been considered a more suitable choice for anomaly detection in time series data [17].

Reconstruction-based techniques have embraced the new generation of deep generative networks, such as Variational Autoencoders (VAE) [14] and Generative Adversarial networks (GAN) [15] that have made possible the task of non-linear dimensionality reduction and learning representation of input data in unsupervised manner. We have given preference in this review by tracing the studies that contribute to the body of works on use of deep generative networks for anomaly detection in time series data.

A Variational Bi-LSTM Autoencoder proposed by Pereira and Silveira (2018) [22] uses a special attention mechanism called Variational SelfAttention Mechanism (VSAM) to improve the decoding process for anomaly detection in solar energy time series data. The VAE-reEncoder architecture, composed of two Bi-LSTM encoders and Bi-LSTM decoder, proposed for detecting anomalies in time-series data by Zhang and Chen (2019) in [19] has also been inspirational for our work. Two scores were taken into account: 1) a reconstruction error of the decoder and 2) a reconstruction error calculated by comparing the latent vectors generated by the second and the first encoder. An and Cho (2015) [23] propose new method for computing anomaly score with reconstruction probability using Variational Autoencoders that motivated our interest in investigating the potential of using the Mahalanobis distance in our comparative analysis.

Of particular interest to the present work are the findings of seven research studies that have participated in the original BATADAL competition against which we compare our results. The approaches taken by the participant teams range from data-driven to model-based. The best-performing model on the challenge task was reported by Housh and Ohar (2017) [24] that have built a simulation model of the hydraulic processes in the water distribution system using the popular EPANET [25] toolkit for modeling water distribution systems. Subsequently, the anomalies were identified by comparing the simulated values with the observed values provided by the dataset.

Abokifa et al. (2017) [26] introduced a three-stage detection method; the first one using statistical methods for detecting outliers, the second using a Multi-layered perceptron targeting contextual anomalies and the third one aimed at detecting global anomalies affecting multiple sensors by employing Principal Component Analysis as a dimensionality reduction method. Drawing on this experiences, we have also explored the idea of detecting contextual anomalies with neural networks, in our case it was a deep learning architecture instead of a Multi-layered Perceptron. The detection method reported in Giacomoni et al. (2017) [27], identifies the values below and above the thresholds associated with normal operation conditions for the water distribution system. Subsequently, Principal Component Analysis was used for the separation of normal data and those that are considered anomalies.

In an attempt to reduce the dimensionality and complexity of the problem, Brentan et al. (2017) [28], propose using an independent deep recursive neural network for each district (a division that was inherent to the C-Town water supply data) to predict the tank water levels as a function of other available features provided in the dataset. We have also explored this idea of independent

detection across all districts and examined the impact on the performance of our models when compared to a more holistic approach of having one model for detecting the cyber-attacks. In another line of work, Chandy et al. (2017) [29] have adopted a two-phase detection model. First, any violations of the physical and/or operating rules are flagged as potential anomalies, which are latter confirmed by the second model, a Convolutional Variational Autoencoder (VAE) that calculates the reconstruction probability of the data. In our approach, we have also opted for the use of deep generative network, VAE, although we use hierarchical stacking of various layers (e.g., convolutional and LSTM) as opposed to a pure convolutional VAE structure that was used in their work.

Pasha et al. (2017) [30] presented an algorithm consisting of three main interconnected modules working on control rules and consistency checks, statistical pattern recognition for hydraulic and system relationships. A two-stage method extracts a four dimensional feature vector to capture the mean and covariance present in the multi-dimensional time series data, and latter uses Random Forest classifier to detect the cyber attacks was reported by Aghashahi et al. (2017) [31].

3 Dataset

The task of detecting cyber-physical attacks on water distribution system was the central objective posed by the competition challenge the BATtle of the Attack Detection ALgorithms (BATADAL) [20]. The accompanying dataset contained simulated observation data from the hydraulic monitoring system of a hypothetical C-Town medium-size water network. The participant were provided with two training and one test dataset, containing time series data sampled hourly from the simulated observations of 43 system variables divided into nine district metered areas.

The first training set contained system observations under normal operating conditions (no cyber attacks) for a period of 365 days. The second training dataset included 7 cyber attacks, one revealed and the other partially revealed or concealed from the participants. Detection algorithms were evaluated on a test dataset containing 7 additional concealed attacks, using the two scores described below.

Time-to-detection, denoted as S_{TTD} , measuring how fast an algorithm detects an attack was calculated as:

$$S_{TTD} = 1 - \frac{1}{N} \sum_{i=1} \frac{TTD_i}{\Delta_i} \quad (1)$$

where, N is the number of attacks, TTD_i is the time to detect (in hours) relative to the i -th attack and Δ_i is duration of an attack.

Classification performance score, S_{CLF} , assessing how accurate an algorithms is, was defined as:

$$S_{CLF} = \frac{TPR + TNR}{2} \quad (2)$$

where, TPR is true positive rate usually known as recall, and TNR is true negative rate usually known as specificity, which were calculated in a standard way based on the TP, TN, FP and FN.

These two evaluation scores were used in an overall ranking score S , defined as:

$$S = y * S_{TTD} + (1 - y) * S_{CLF} \quad (3)$$

The value of the coefficient was set to 0.5, giving equal weights to both evaluation metrics i.e. early and accurate detections.

We have performed some feature preparation by adding seasonal factors encoding derived from timestamps, such as sin and cos value of month-of-year, day-of-month, and hour-of-day; min-max normalization was subsequently performed for each seasonal dimension.

4 Methodology

4.1 Variational Autoencoders for Anomaly Detection

When faced with multivariate time series data, such as the BATADAL dataset, and being inclined toward exploring the potential of deep learning for anomaly detection, Variational Autoencoders [14] seemed an obvious choice. We give a brief overview of general VAE architecture shown in Fig. 1. Variational Autoencoder consists of two neural networks, encoder and decoder. The encoder maps high-dimensional input data $\mathbf{x}$ into a low-dimensional latent space $\mathbf{z}$, while the role of the second neural network, decoder, is to generate or reconstruct the input data from the sampled latent vector $\mathbf{z}$ as reconstructed input data $\mathbf{x}'$.

Calculating the posterior distribution $p(\mathbf{z} | \mathbf{x})$ is not tractable, so variational inference is used to approximate p with some variational distribution $q(\mathbf{z} | \mathbf{x})$; usually Gaussian distribution so that it can be feasibly calculated. Kullback-Leibler divergence (D_{KL}) is most commonly used to measure how close is one distribution to another, and in the case of VAE, it measures how close is the approximative posterior distribution $q(\mathbf{z} | \mathbf{x})$ to the true prior $p(\mathbf{z})$. For calculation purposes, log-likelihood formulas are used, hence instead of minimizing $D_{KL}(q(\mathbf{z} | \mathbf{x}) || p(\mathbf{z} | \mathbf{x}))$, the evidence lower bound ELBO, which is the lower bound of log-likelihood distribution of input data $p(\mathbf{x})$ is maximized.

Variational Autoencoders try to maximize a stochastic objective function on training data given by the following equation:

$$ELBO(\theta, \psi; x) = E_{q_\theta(\mathbf{z} | \mathbf{x})} \log(p_\psi(\mathbf{x} | \mathbf{z})) - D_{KL}(q_\theta(\mathbf{z} | \mathbf{x}) || p(\mathbf{z})) \quad (4)$$

The first term, represents the reconstruction likelihood i.e. how good is the generative model at learning the training data, where $p_\psi(\mathbf{x} | \mathbf{z})$ denotes the likelihood of obtaining the input vector $\mathbf{x}$ given the latent vector $\mathbf{z}$, parametrized by the decoder neural network. The second term is the Kullback-Leibler divergence with a prior over the latent variables, where $q_\theta(\mathbf{z} | \mathbf{x})$ is the approximated posterior distribution parameterized by the encoder neural network.

We estimate ELBO by sampling from q , so the “re-parametrization trick” in VAE is used. The encoder neural network that needs to learn the approximated posterior distribution $q(\mathbf{z} | \mathbf{x})$ takes the input $\mathbf{x}$ and outputs the parameters of $q(\mathbf{z} | \mathbf{x})$, e.g., a vector of means $\boldsymbol{\mu}$ and a vector of standard deviations $\boldsymbol{\sigma}$ for Gaussian distribution, from which $\mathbf{z}$ is sampled as an element-wise product of $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ and passed to the decoder (see Fig. 1).

During training epochs, the first term of the objective function i.e., reconstruction likelihood, is calculated either in a traditional way as a mean square error or a Mahalanobis distance. Mahalanobis distance [21] takes into account the covariance between each pair of variables that exist in the space, therefore providing much more realistic and reliable way of measuring a distance between two points in a multivariate space. The equation for Mahalanobis distance between two points (X_a and X_b) that exists in n dimensional space with (nonsingular) covariance matrix Σ is the following:

$$Mh_{ab} = \sqrt{(X_b - X_a)^T * \Sigma^{-1} * (X_b - X_a)} \quad (5)$$

In our models, the Mahalanobis distance was computed between the original input vector $\mathbf{x}$ and the reconstructed input vector $\mathbf{x}'$ (averaged over 5 sampled vectors).

The anomaly threshold for every instance at time t is determined as follows:

$$|Mh_{xt}x'_t - \mu_n| > 3 * \sigma_n \quad (6)$$

where, $\mathbf{x}_t$ is the input vector at time t and $\mathbf{x}'_t$ is the averaged reconstructed input, μ_n and σ_n are the mean and standard deviation of the computed Mahalanobis distances for the previous n timestamps.

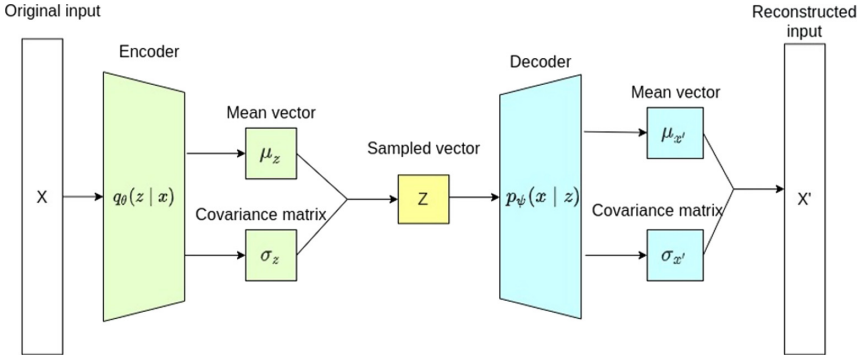


Fig. 1. Variational Autoencoder architecture

4.2 Models

Motivated by previous methodologies evaluated on the same dataset [26, 28, 29] that take into account the natural division of the C-Town water network into nine district metered areas, we have taken two modelling approaches. The first modelling approach takes a holistic view of the water distribution system operations. Modeling each of the nine district metered areas independently was used as an alternative approach, allowing us to reduce the dimensionality of the system time series data into nine components. In the second approach, the anomalies were detected per component and subsequently merged into a single set that was evaluated by the competition system. We have trained and tested several variants of Variational Autoencoders, the design decision were guided by previous research in the field. Each model has its complement, one modeling the entire water distribution system and the other modeling each time series originating from different district metered area – the area-based approach will be denoted by the letter C during acronym naming.

A short description of each of the ten architectures follows:

VAE-D. The encoder and decoder of this VAE variant are two neural networks, each represented by one dense layer. Each layer has 18 neurons, uses ReLU activation function and 5-dimensional latent space.

VAE-D-C. The simple architecture of VAE-D with two neural networks with dense layers was created for each time series component (corresponding to one district metered area). The number of units per layer and the dimensionality of the latent space were tailored to the specifics of the district metric area being modeled.

LSTM-VAE. This VAE variant has two LSTM networks as encoder and decoder in an attempt to capture the long-term dependencies when modeling sequences. The encoder is consisted of 1 LSTM layer with 18 units with ReLU connected to Dense layer with number of units to parametrize the 6 dim latent space. The input to the model is a window of 24 consecutive hours of data.

LSTM-VAE-C. This is a variant that uses one LSTM-VAE for each district metered area i.e., each of the 9 time series components.

LSTM-VAE-2E. Variational AutoEncoder with two encoders and a decoder. The first encoder has the simple architecture of VAE-D that generates the sampled encodings in the latent space. The second encoder is an LSTM neural network with two stacked layers that learns to predict an instance for the next timestamp given a sequence of previous 24 instances; the second LSTM layer outputs the final hidden state for the sequence. This final state is taken as a compression where all the relevant information are kept for the sequence in order to predict the next instance in the time series. Each LSTM layer has 10 neurons with activation ReLU and uses dropout. The input of these two encoders is concatenated and fed into a decoder. There is single model for the time series.

LSTM-VAE-2E-C. Same architecture as previously described LSTM-VAE-2E is used to model each of the 9 components, by adjusting the hyperparameters to the modeled system parameters.

CNN-VAE. Guided by the modeling approach taken in [29], we have explored the performance of a VAE variant that has an encoder, which is a Convolutional neural network (CNN) suitable to extract the local contextual relationships between the instances in our time series data. We use CNN with one convolution layer, Max pooling with filter size of 27 and kernel size of 6. The decoder has the same architecture as VAE-D.

CNN-VAE-C. This is component-wise variant of the previous model CNN-VAE, i.e., 9 CNN-VAE networks for each area with their hyperparameters selected to fit the problem being modeled.

VAE-ReEncoder. In this model, similarly as in [19], we use the encoder twice. In the first pass the encoder generates latent distribution from the input data that is sampled and passed through the decoder, so that in the second pass, the output generated by the decoder is sampled and fed to the input of the encoder. Two scores are taken into account for computing the anomaly score: 1) the Mahalanobis distance between the sampled reconstructed vector $\mathbf{x}'$ and the input vector $\mathbf{x}$ and 2) the mean square error between the latent vectors generated by the first and second pass through the encoder.

VAE-ReEncoder-C. Same as VAE-ReEncoder but here we have a different model for each component.

5 Discussion of Results

An exploratory study was conducted to assess the performance standing of ten hierarchical Variational Autoencoders architectures trained and evaluated on BATADAL datasets. The same evaluation criteria established for the BATADAL competition were used to compare the performance of the models, namely, the best overall score (S), time-to-detection (S_{TTD}) and classification performance score (S_{CLF}). Table 1 presents the evaluation metrics for each VAE model using the proposed Mahalanobis metric as a reconstruction loss. Surprisingly, the first model with the most simple architecture, VAE-D has obtained the best results on two metrics, overall score and the classification performance score. It seems that the model was able to capture the relationships between the 9 system components (areas), since its complemented architecture VAE-D-C that models each component separately underperforms on the same task.

The current findings demonstrate lower performance achieved by the LSTM variants of Variational Autoencoders, LSTM-VAE and LSTM-VAE-C, when compared with the best performing model. However, both of the models has shown almost perfect time-to-tetection score with LSTM -VAE-C being the model with the best TPR i.e. recall score. But the LSTM-VAE has failed to detect enough anomalies and yielded poor recall performances and LSTM-VAE-C has shown poor TNR i.e. specificity performances. We could hypothesized

Table 1. Evaluation metrics of the proposed deep learning models using Mahalanobis distance as a distance metric

Model	S	S_{TTD}	S_{CLF}	Precision	#Attacks detected
VAE-D	0.800	0.975	0.624	0.412	7
LSTM-VAE	0.735	0.979	0.491	0.185	7
LSTM-VAE-2E	0.664	0.820	0.508	0.236	7
CNN-VAE	0.523	0.543	0.503	0.227	5
VAE-ReEncoder	0.752	0.935	0.570	0.276	7
VAE-D-C	0.778	0.987	0.570	0.257	7
LSTM -VAE-C	0.778	0.999	0.556	0.216	7
LSTM-VAE-2E-C	0.761	1.0	0.523	0.501	7
CNN-VAE-C	0.713	0.931	0.496	0.191	7
VAE-ReEncoder-C	0.726	0.940	0.512	0.211	7

that the reason for the results lies in the complexity of the networks i.e., the dataset was not large enough to learn the high number of parameters needed to be learned.

The LSTM-based VAEs with 2 encoders have also shown lower performances, while the LSTM-VAE-2E-C being the model with best precision. The result could be contributed to the use of two encoders, the first one tasked with extracting high-dimensional features and relationship from the normal input observations, while the other encoder, was capable of extracting the temporal dependencies in a particular window of data using LSTM networks. The lack of large enough training set still accounts for lower performance compared with the simplest and best performing model VAE-D.

Little or no evidence has supported the advantages of using Convolutional variants of VAEs for cyberattack detection on the water distribution system, although CNN-VAE has yielded best TNR, but at the same time very poor score for TPR. One possible speculation for the results, could be attributed to the inadequacy between the size of the training dataset and the number of parameters needed to be learned. In order to capture all local and temporal dependencies from high dimensional time series the convolution layers needed better tuning. Meanwhile, the Conv-VAE-C version that independently models each district area, has shown results closer to the others, more likely because the convolutional networks were with lesser complexity i.e., the parameters were trained on partitions of the time series data.

The VAE-ReEncoder variants have similarly exhibited unsatisfactory performances. The reason could be the one previously discussed, the performance gains of deep learning networks are notoriously dependent on the amount of training data. In addition, more experiments for hyperparameters tuning are required to further advance the architectures.

One of the objective of this study, was to evaluate the impact of using Mahalanobis distance as opposed to the traditional distance metric, such as mean square error. We have conducted additional experiments for comparative analysis of the models and the results are displayed in Table 2. As hypothesized, the evaluation of the best performing models, VAE-D and VAE-D-C, revealed the significance of using a metric that is most suited to the task. It can be seen that both models have shown significant performance advantage when Mahalanobis distance is used, pointing to the suitability and relevance of using more appropriate measure to represent the loss in the distribution output of VAEs as well as a metric that takes into account the variance that exists in multivariate data.

Table 2. Comparison of the evaluation metrics of the best performing models using mean square error and Mahalanobis distance as a distance metric

Model	S	S_{TTD}	S_{CLF}	Precision	#Attacks detected
VAE-D	0.800	0.975	0.624	0.412	7
VAE-D-MSE	0.702	0.821	0.584	0.533	7
VAE-D-C	0.778	0.987	0.570	0.257	7
VAE-D-C-MSE	0.750	1.0	0.500	0.194	7

Table 3. Comparison of the evaluation metrics of our best performing model VAE-D with the models submitted to the BATADAL competition task

Rank	Team	#Attacks detected	S	S_{TTD}	S_{CLF}
1	Housh and Ohar	7	0.970	0.975	0.953
2	Abokifa et al.	7	0.949	0.958	0.940
3	Giacomoni et al.	7	0.927	0.936	0.917
4	Brentan et al.	6	0.894	0.857	0.931
5	Chandy et al.	7	0.802	0.835	0.768
6	VAE-D	7	0.800	0.975	0.624
7	Pasha et al.	7	0.773	0.885	0.660
8	Aghashahi et al.	3	0.534	0.429	0.640

Table 3 presents the performance results of our best model and the results reported by the 7 participant teams at the BATADAL competition. The superiority of our model in terms of time to detection of an attack is evident, although the VAE-D models fails to prevent many false positives and to detect all the instances that belong to an attack. It is interesting that it almost perfectly detects if there is an attack or in other words with almost perfect latency it detects the first instance of a series of instances during an attack, but it fails to

detect the following instances in the ongoing attack. We would like to underline that no validation dataset was used for detecting anomalies, which might be a possible explanation for the lower performance compared to the best performing BATADAL models. We defer the discussion on the impact the cut-off thresholds for defining outliers has on the performance, until a detailed analysis is done.

6 Conclusion and Future Work

The present research explores several variants of Variational Autoencoders on the task of detection of attacks on a simulated observations in a water distribution system. The premise of the proposed methods for unsupervised deep learning anomaly detection is that attacks i.e., anomalies are events that cannot fit perfectly with the model operating under normal conditions. Variational Autoencoders use two neural networks, encoder and decoder, to model and learn the normal data. Our best performing model showed almost perfect results in time to detection of intrusions that are sparse and never encountered before without using any domain-specific knowledge during training. The effect of using the Mahalanobis distance as a reconstruction error is promising although needs further investigation. Future research directions will focus on increasing the model performance standing towards better recall and sensitivity. We continue to experiment with hyperparameter tuning and transfer learning on various benchmark datasets. Investigations of advanced techniques, such as attention mechanism and self-supervised learning will also be explored. Taken together, our exploration study fits into the line of research on deep generative learning for anomaly detection.

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